

John D. (Jack) Treado

POSTDOCTORAL RESEARCHER

Cluster of Excellence Physics of Life, TU Dresden

Max Planck Institute for the Physics of Complex Systems

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EDUCATION

Ph.D. in Engineering & Applied Sciences

Yale University

Advisor: Prof. Corey O'Hern

2016–2022

New Haven, CT, USA

B.Sc. in Physics, *magna cum laude*

Georgetown University

Bachelor's Thesis Advisor: Prof. Peter Olmsted

2012–2016

Washington, DC, USA

ACADEMIC APPOINTMENTS

Postdoctoral Researcher

Cluster of Excellence Physics of Life, TU Dresden

Advisor: Prof. Dr. Otger Campàs

2022–present

Dresden, Germany

Visiting Postdoctoral Researcher

Biological Physics Group, Max Planck Institute for the Physics of Complex Systems

Advisor: Prof. Dr. Frank Jülicher

2022–present

Dresden, Germany

SELECTED PUBLICATIONS

1. **JDT**, A. Boutillon, F. Jülicher, O. Campàs. "Hydraulics of Active Tissues." *in preparation* (2026).
2. I. Di Terlizzi*, L. Koehler*, and **JDT***. "Passive memory reshapes active persistence," [arXiv:2605.29162](#) (2026).
3. **JDT**, A. B. Roddy, G. Thérout-Rancourt, C. Ambrose, C. Brodersen, M. D. Shattuck, and C. S. O'Hern, "Localized growth and remodelling drives spongy mesophyll morphogenesis," *J. R. Soc. Interface* **19**, 20220602 (2022).
4. **JDT***, D. Wang*, A. Boromand, M. P. Murrell, M. D. Shattuck, and C. S. O'Hern, "Bridging particle deformability and collective response in soft solids," *Phys. Rev. Materials* **5**, 055605 (2021).
5. Z. Mei*, **JDT***, A. T. Grigas, Z. A. Levine, L. Regan, and C. S. O'Hern, "Analyses of protein cores reveal fundamental differences between solution and crystal structures," *Proteins: Structure, Function, and Bioinformatics* **88**, 1154 (2020).
6. **JDT**, Z. Mei, L. Regan, and C. S. O'Hern, "Void distributions reveal structural link between jammed packings and protein cores," *Phys. Rev. E* **99**, 022416 (2019).

* denotes equal contribution

GRANTS & FELLOWSHIPS

Yale Mechanical Engineering & Materials Science Goodyear Tire & Rubber Fellow

2020

SELECTED TALKS & PRESENTATIONS

Invited Talk , Interdisciplinary Challenges in the Physics of Complexity and Life 2027, <i>Padova, Italy</i>	2027
Seminar , Laboratoire Physique des Cellules et Cancer, Institut Curie, <i>Paris, France</i>	2026
Seminar , James Franck Institute, University of Chicago, <i>Chicago, IL, USA</i>	2025
Seminar , Department of Physics and Astronomy, University of Padova, <i>Padova, Italy</i>	2025
Invited Talk , WWTF Annual Meeting, Universität Wien, <i>Vienna, Austria</i>	2022
Seminar , Physics of Living Systems (PoLS) Seminar Series, <i>Virtual</i>	2021
Invited Talk , 4th International Conference on Packing Problems, <i>New Haven, CT</i>	2019

TEACHING & MENTORSHIP EXPERIENCE

Master's Thesis Supervisor <i>Cluster of Excellence Physics of Life, TU Dresden</i>	2025-2026 Dresden, Germany
Teaching Fellow — Research Methods in Physics and Biology (PHYS 099) <i>Department of Physics, Yale University</i>	Spring 2020 - 2021 New Haven, CT, USA
Teaching Fellow — Mechanical Engineering III: Dynamics (MENG 383) <i>Department of Mechanical Engineering & Materials Science, Yale University</i>	Fall 2020 New Haven, CT, USA
Teaching Fellow — Computing for Engineers and Scientists (ENAS 130) <i>Department of Mechanical Engineering & Materials Science, Yale University</i>	Spring 2019 New Haven, CT, USA
Teaching Fellow — Integrated Workshop (ENAS 991) <i>Department of Mechanical Engineering & Materials Science, Yale University</i>	Fall 2017 - 2019 New Haven, CT, USA

AWARDS & HONORS

Excellence in Poster Presentation, Granular Matter Gordon Research Seminar, <i>Stonehill College, MA</i>	2018
Georgetown University Physics Department Undergraduate Research Award	2016
Georgetown University Physics Department Paul A. Treado Medal	2016
Georgetown University Physics Department Departmental Honors	2016

PROFESSIONAL SERVICE

Organizer, Interdisciplinary Challenges in the Physics of Complexity and Life 2028, <i>Marseille, France</i>	2028
Postdoctoral Representative — <i>Cluster of Excellence Physics of Life, TU Dresden</i>	2026
Postdoctoral Representative — <i>MPI-PKS</i>	2023-2025
Public science publication, "Protein Folding: Nature's Rubik's Cube," <i>Hartford Courant</i>	2019
Public lecture, "Finding Patterns in Chaos: How Simple Rules Form Complex Behaviors," <i>Yale Science Diplomats</i> .	2018